Predicting Delivery Delays

Using Operational Logistics Data

Group 7 | Capstone Project

Mentor

Gaurav Chauhan

Team Members

Roshni | Sashank Bhargava | Amal Krishna A | Sri Shewatha K

Machine Learning · Logistics Analytics · Delay Prediction

Problem Definition

BUSINESS PROBLEM

E-commerce logistics companies handle millions of shipments daily. A key operational challenge is **unpredictable delivery delays**, which lead to:

- Customer dissatisfaction
- Increased support costs
- Operational inefficiencies
- Loss of brand trust

Logistics teams typically identify delays only after they occur, leaving no room for proactive action.

TECHNOLOGY PROBLEM

Develop a machine learning model that predicts whether a shipment will be delayed using:

- Shipment characteristics
- Courier partner details
- Geographic delivery zones
- Operational parameters

Goal

Enable proactive intervention before delays happen — shifting from reactive to predictive logistics operations.

Why This Matters

250K+
Shipment records in dataset

~20%
Shipments are delayed

4×
Higher cost of post-delay handling vs. prevention

Business Impact of Delivery Delays

Customer Churn
Late deliveries erode trust on e-commerce platforms, driving customers to competitors.

RTO Costs
Return-to-origin shipments from failed deliveries inflate operational expenses.

Courier Evaluation
Without predictive data, poor-performing couriers go undetected until damage is done.

Logistics Planning
Unpredictable delays prevent efficient resource allocation and route optimization.

Value Additions Planned

1 Advanced Feature Engineering

Created meaningful operational features capturing real logistics constraints:

Shipment Volume	Aggregate load at pickup/drop
Chargeable Weight Proxy	Volumetric weight estimation
Courier × Zone Interaction	Performance by region
State Lane	Origin–destination corridor
Promise Window	Delivery tightness metric

2 Risk-Based Delay Prediction

Instead of predicting delivery days directly, we predict the probability of delay — aligning better with operational decision-making.

Traditional Approach

Predict exact delivery time
Binary pass/fail after the fact
No actionable lead time

Our Approach

Predict probability of delay before shipment is dispatched
Enables proactive intervention

Result: high-risk shipments flagged early for prioritization, courier reassignment, or adjusted commitments.

Suggested Solution & Dataset

ML-Based Delay Prediction System

The model uses operational shipment data to estimate the probability of a delivery delay, enabling logistics teams to act before delays occur.

1. **Ingest**
   Operational shipment data from 250K+ records
2. **Engineer**
   Derive features: volume, density, courier-zone interactions
3. **Predict**
   ML model outputs delay probability per shipment
4. **Act**
   Flag high-risk shipments for proactive intervention

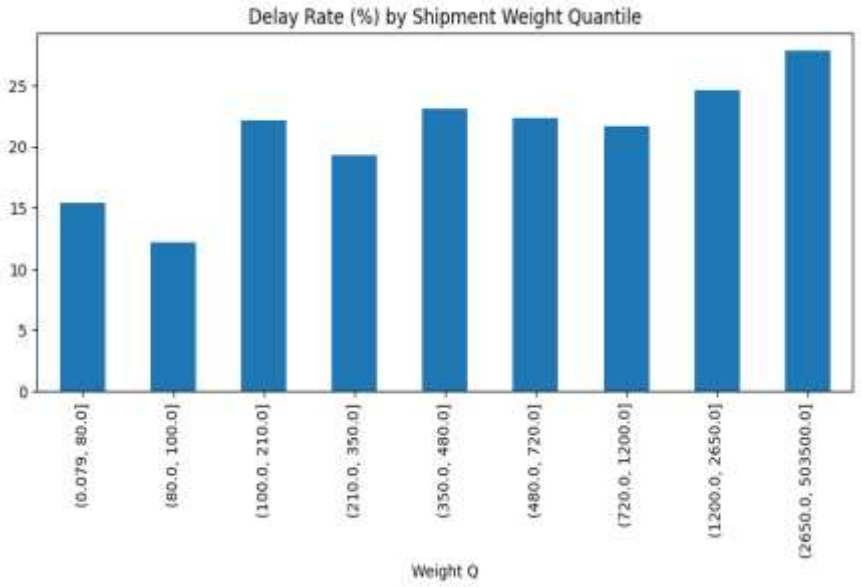
Dataset Overview

250K+	Shipment records
~20%	Delay rate (class imbalance)
10+	Raw features used
5+	Engineered features added

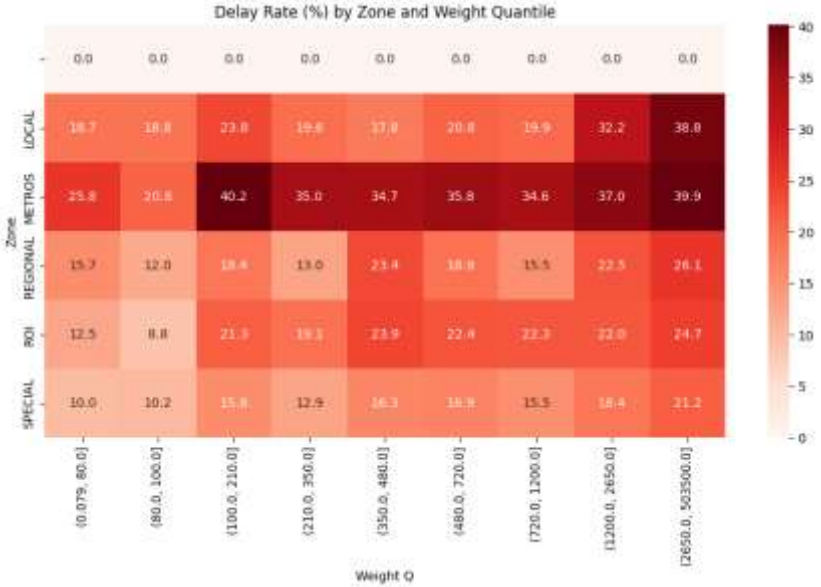
Key fields: shipment weight, dimensions, courier partner, pickup/delivery location, geographic zone, payment mode, delivery commitment windows (Min/Max TAT)

EDA: Shipment Weight & Zone Analysis

Delay Rate (%) by Shipment Weight Quantile



Delay Rate (%) by Zone & Weight Quantile



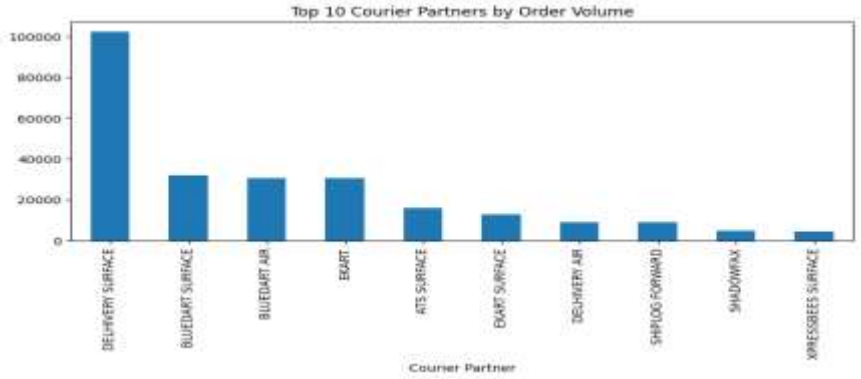
▲ Heavier shipments (>1200g) show 25–28% delay rates vs. 12% for mid-range weights

▲ METROS zone consistently shows highest delays (35–40%) across all weight bands

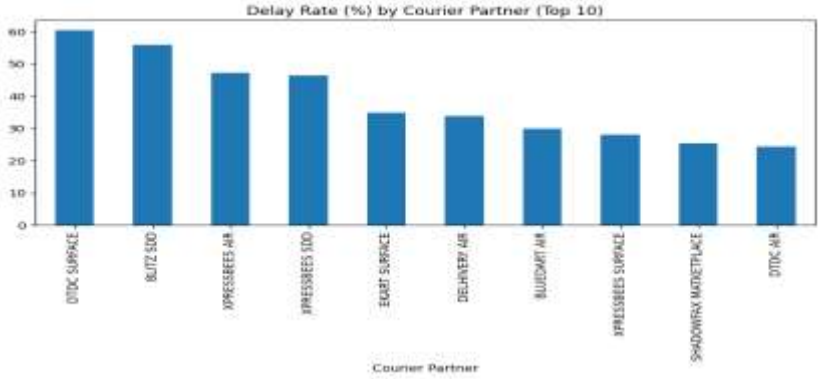
▼ SPECIAL & ROI zones show lowest delay rates — better infrastructure coverage

EDA: Courier Partner Analysis

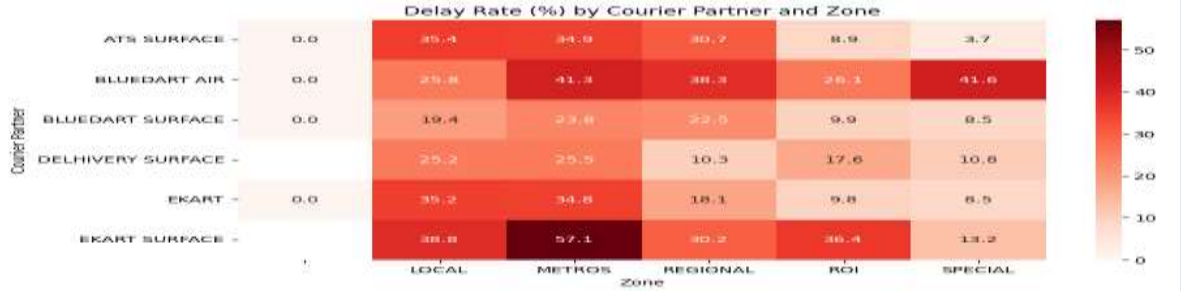
Top 10 Couriers by Order Volume



Delay Rate (%) by Courier Partner — Top 10



Delay Rate (%) by Courier Partner & Zone

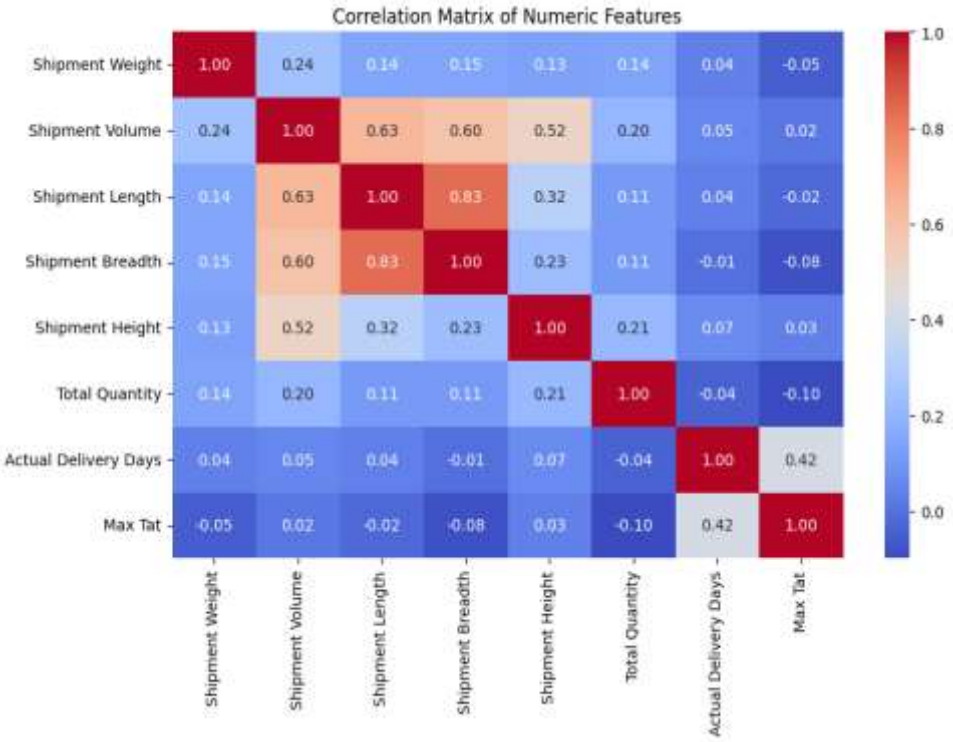


Key Findings

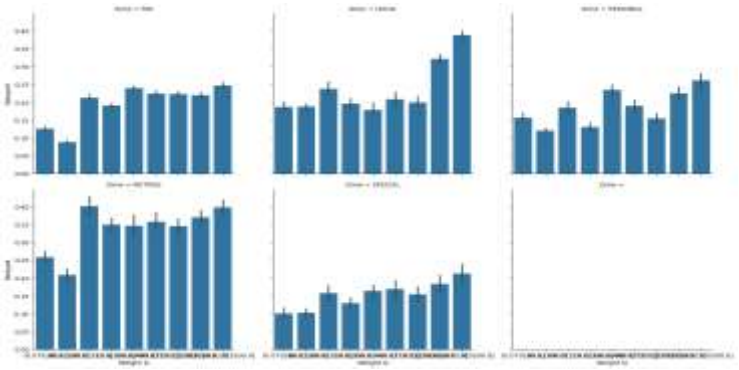
- DTDC Surface & Blitz SDD have 55–60% delay rates
- EKART SURFACE: 57% delay rate in METROS zone
- Delhivery Surface dominates volume but has moderate delays (~25%)
- Bluedart Air shows high delays in METROS & SPECIAL zones

EDA: Feature Correlations & Zone-Level Patterns

Correlation Matrix of Numeric Features



Delay Rate by Zone & Weight (with CI)



Length–Breadth

r = 0.83

High collinearity — only one needed as feature

Volume–Length/Breadth

r = 0.60–0.63

Volume already captures these dimensions

Actual Days–Max TAT

r = 0.42

Delivery time is tied to committed window

Weight → Delay

r = Low direct

Weight effect is indirect — captured via engineered features

Data Challenges & How We Addressed Them

Class Imbalance

Challenge: Only ~20% of shipments were delayed

Solution:

Careful model evaluation and threshold tuning rather than accuracy-only metrics

High Cardinality Variables

Challenge: Courier partners & channels had many categories

Solution:

Appropriate encoding and feature engineering; CatBoost's native handling

Missing & Incomplete Data

Challenge: Non-completed shipments lacked delivery dates

Solution:

Incomplete shipments were excluded because delay cannot be determined without a final delivery date

Data Leakage Risk

Challenge: Delivery outcome variables correlated with target

Solution:

Excluded post-delivery variables from training features to ensure realistic predictions

Algorithms Considered

Baseline
Logistic Regression

STRENGTHS

- Simple & interpretable
- Fast to train
- Good for comparison

LIMITATIONS

- Limited nonlinear relationships
- Lower performance vs. ensembles

Ensemble
Random Forest

STRENGTHS

- Handles nonlinear data
- Robust to noise/overfitting
- Works with mixed features

LIMITATIONS

- Less interpretable
- Needs feature encoding

Gradient Boost
XGBoost

STRENGTHS

- Strong predictive power
- Handles complex interactions
- Efficient framework

LIMITATIONS

- Needs hyperparameter tuning
- Can overfit if not regularized

Best Model ★
CatBoost

STRENGTHS

- Native categorical handling
- Strong on structured data
- Less encoding needed

LIMITATIONS

- Higher compute cost
- GPU recommended for scale

Experiment: Courier Partner Clustering

Motivation

Couriers showed different delivery performance patterns in EDA. Could grouping them into behavioral clusters improve delay prediction?

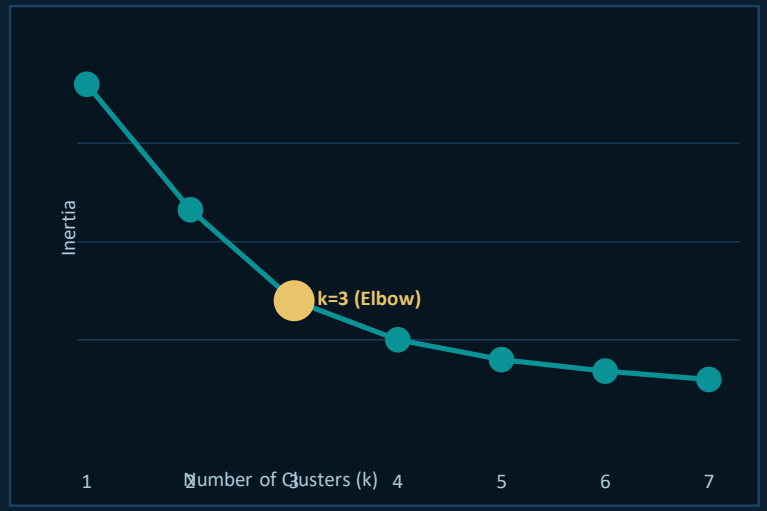
Approach

Applied K-Means clustering on courier-level stats: delay rate, average delivery time, shipment volume, route characteristics. Evaluated using Elbow method and Silhouette score.

Outcome

Distinct courier segments found, but cluster labels did not significantly improve model performance. Original courier feature already captured most signal.

Elbow Method for Cluster Selection



Courier Segments Identified

- High Reliability
- Moderate Reliability
- High Delay Risk

Note: Cluster labels were not included in the final model as they did not improve performance.

Model Performance Results

For delay prediction, **Recall is the priority metric** — a missed delay (false negative) is far costlier than a false alarm.

CatBoost: Recall Trade-off Across Tuning Phases

Phase 1

0.711

RECALL

AUC: 0.808
Precision: 0.414

VS

Phase 3 (GridSearchCV)

0.620

RECALL

AUC: 0.808
Precision: 0.460

Same AUC (0.808), but Phase 1 CatBoost catches **9% more actual delays**. Tuning optimised for F1, shifting the boundary toward precision.

All Models — Phase 1 Results

Model	AUC	Recall	Precision
CatBoost ★	0.808	0.711	0.414
XGBoost	0.800	0.735	0.396
Neural Net	0.796	0.266	0.647
Random Forest	0.753	0.753	0.328

Computational Challenge
GridSearchCV on CPU exceeded 4 hrs and was cancelled. Resolved by **switching to Nvidia GPU + reducing parameter grid**.

Deployment: Phase 3 CatBoost with a lower threshold (< 0.5) recovers high recall without retraining.

Phase 2 Results — RandomSearchCV

Hyperparameter tuning with RandomSearchCV — models ranked by **F1 score**. XGBoost leads with AUC 0.806 and best F1 0.531.

Model	ROC AUC	Precision	Recall	F1
XGBoost ★	**0.806**	**0.460**	**0.628**	**0.531**
CatBoost	0.806	0.453	0.633	0.528
Random Forest	0.799	0.457	0.611	0.523
Neural Net (MLP)	0.788	0.426	0.636	0.510

XGBoost tops Phase 2 on F1 (0.531). All models show **improved recall vs Phase 1** (RandomSearchCV optimised for F1 rather than recall alone).

Phase 3 Results — GridSearchCV

Fine-tuned with GridSearchCV — CatBoost reclaims top spot with **AUC 0.808**. Neural Net achieves highest recall (0.64) in this phase.

Model	ROC AUC	Precision	Recall	F1
CatBoost ★	**0.808**	**0.46**	**0.62**	**0.53**
XGBoost	0.805	0.46	0.62	0.53
Random Forest	0.796	0.47	0.60	0.52
Neural Net (MLP)	0.793	0.44	0.64	0.52

CatBoost leads with highest AUC (0.808). Deployment recommendation: use Phase 3 CatBoost with **threshold < 0.5** to recover high recall without retraining.

Cross-Phase Comparison: Best Model per Phase

How tuning evolved from baseline to GridSearchCV — balancing AUC, Recall and F1

Phase	Best Model	ROC AUC	Precision	Recall	F1
Phase 1	CatBoost	**0.808**	0.414	0.711	—
Phase 2 (Random)	XGBoost	**0.806**	0.460	0.628	**0.531**
Phase 3 (Grid)	CatBoost	**0.808**	0.460	0.620	**0.530**

Phase 1 — Baseline
CatBoost leads. High recall (0.711) but low precision (0.414). No F1 metric tracked.

Phase 2 — RandomSearchCV
XGBoost tops on F1 (0.531). Better balance across all metrics vs Phase 1.

Phase 3 — GridSearchCV ★
CatBoost best AUC (0.808). Lower threshold recommended to boost recall in deployment.

Key Takeaways & Conclusions

01 Prediction is Feasible

CatBoost achieved ROC AUC of 0.808 — a meaningful predictive signal from operational data alone, enabling proactive delay management.

02 Feature Engineering Was Critical

Derived features (courier-zone interaction, chargeable weight, promise window tightness) significantly improved signal quality beyond raw data.

03 Geography & Couriers Drive Delays

Metro zones and specific courier-zone combinations are the primary delay predictors — actionable insight for logistics network planning.

04 Threshold Tuning for Deployment

Phase 1 CatBoost catches 9% more delays at the same AUC. Recommended: use Phase 3 weights with a lower classification threshold (< 0.5) to dial recall up without retraining.

Thank You

Predicting Delivery Delays Using Operational Logistics Data

Group 7 | Roshni · Sashank Bhargava · Amal Krishna A · Sri Shewatha K

Special Acknowledgments to Ashvini Jhakar (CEO, Prozo Technology), Abhishek Saini (Analyst, Prozo), Mentor Gaurav Chauhan, and the AI tools ChatGPT and Claude AI for their valuable support and guidance.